

LSTM-based aerodynamic force modeling for unsteady flows around structures

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Abstract. The aerodynamic force is a significant component that influences the stability and safety of structures. It has unstable properties and depends on computer precision, making its long-term prediction challenging. Accurately estimating the aerodynamic traits of structures is critical for structural design and vibration control. This paper establishes an unsteady aerodynamic time series prediction model using Long Short-Term Memory (LSTM) network. The unsteady aerodynamic force under varied Reynolds number and angles of attack is predicted by the LSTM model. The input of the model is the aerodynamic coefficients of the 1 to n sample points and output is the aerodynamic coefficients of the n+1 sample point. The model is predicted by interpolation and extrapolation utilizing Unsteady Reynolds-average Navier-Stokes (URANS) simulation data of flow around a circular cylinder, square cylinder and airfoil. The results illustrate that the trajectories of the LSTM prediction results and URANS outcomes are largely consistent with time. The mean relative error between the forecast results and the original results is less than 6%. Therefore, our technique has a prospective application in unsteady aerodynamic force prediction of structures and can give technical assistance for engineering applications.

Keywords: angles of attack; deep learning; engineering structures; long short-term memory; Reynolds number; unsteady aerodynamic force

1. Introduction

The trend toward lightweight and flexible design has been observed in engineering constructions such as large-span bridges (Jing *et al.* 2018), high-rise skyscrapers (Kwok *et al.* 2009, Cheng *et al.* 2015), and wind turbines (Lienard *et al.* 2019). However, these structures are susceptible to wind-induced vibration damage. This damage is closely associated with complex flow phenomena, including the boundary layer, turbulence, and unsteady features (Szechenyi 1975). It is imperative to study the aerodynamic characteristics of structures to ensure their stability and durability. Consequently, accurately predicting the aerodynamic properties of structures is crucial for structural design and vibration control.

In recent years, wind tunnel testing (Ma *et al.* 2019, Huang *et al.* 2020, Sun *et al.* 2023) and numerical simulation (Blocken 2014, Stringer *et al.* 2014, Bertani *et*

al. 2022, Zhang *et al.* 2023) have been widely used for studying the structural aerodynamic characteristics. Wind tunnel testing has been valuable in accumulating a substantial amount of data in this field. However, due to the complex nature of unsteady aerodynamics, specific tests are still required for structures with different Reynolds numbers and angles of attack (Hu and Kwok 2020). On the other hand, with the improvement of computer performance, numerical simulation methods have gained prominence as an important tool for aerodynamic investigation. These methods allow researchers to control various parameters and conditions during simulations. The Navier-Stokes (N-S) equation is commonly employed in numerical simulations to simulate the flow field. Nevertheless, one of the ongoing challenges in this approach is finding a balance between computational cost and accuracy, especially in complex engineering design scenarios. (Zhang *et al.* 2019).

With the advancement of computer technology and the availability of large datasets, machine learning, particularly deep learning (Lecun *et al.* 2015, Kutz 2017), has found widespread applications in various engineering domains, including structural health monitoring, building materials, and wind energy (Carassale *et al.* 2007, Pathak *et al.* 2018, Wu and Snaiki 2022). Machine learning techniques excel at addressing high-dimensional, complex, and nonlinear problems (Laima *et al.* 2023, Zhang *et al.* 2023). For example, Li *et al.* (2017) conducted cluster analysis on vortex data from long-span bridges, revealing the relationship between velocity fields and vortex modes.

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Huang *et al.* (2017) developed a model for predicting surface wind pressure on high-rise buildings, leveraging a combination of backpropagation neural networks and proper orthogonal decomposition (POD) methods, which significantly enhanced the efficiency of wind tunnel tests. Jin *et al.* (2018) proposed a data-driven fused convolutional neural network model that utilized surrounding pressure data to forecast wind speed, achieving good agreement with numerical simulations. Hu and Kwok (2020) employed various machine learning methods to predict wind pressures around circular cylinders under different Reynolds numbers and turbulence intensities. Wang and Wu (2020) introduced a knowledge-enhanced deep neural network to predict the nonlinear response of structures, resulting in improved computational efficiency without compromising accuracy. Zhai *et al.* (2022) developed a convolutional neural network model capable of effectively detecting cracks. These studies demonstrate the successful utilization of machine learning methodologies in system modeling, damage identification, and load prediction due to their remarkable capability for capturing nonlinear relationships. The application of machine learning techniques in these engineering domains has significantly improved the understanding and analysis of complex phenomena, thereby enhancing the efficiency and accuracy of various engineering processes.

The unsteady aerodynamic force of structures is a nonlinear dynamic problem, and its aerodynamic characteristics are challenging to predict over extended time periods. The powerful machine learning algorithm provides the possibility of unsteady aerodynamic time series modeling. A recent study focus is on exploiting existing high-fidelity data to forecast subsequent time series aerodynamic force.

Recurrent Neural Networks (RNNs) (Rumelhart *et al.* 1986) are a recursive network that have been widely used in natural language processing (NLP). However, RNNs often face issues such as gradient explosion and vanishing for sequences with long-term dependencies (Bengio *et al.* 1994). The long short-term memory (LSTM) network (Hochreiter and Schmidhuber 1997) is designed to overcome the drawbacks of RNNs. LSTM replaces traditional neurons in RNNs with cell units, which solve the gradient explosion and vanishing problems. Therefore, LSTM has been widely used in many fields such as NLP (Ryu *et al.* 2017) and time series forecasting (Yang *et al.* 2019).

LSTM has been widely employed for predicting and modeling unsteady flows. Li *et al.* (2020) developed a reduced-order model of wind-bridge coupling using LSTM to capture the interaction between bridge deck motion and the resulting aerodynamic force. Additionally, Li *et al.* (2020) incorporated the Newmark- β method into LSTM for accurate prediction of bridge flutter response. In a similar vein, Laima *et al.* (2023) utilized bidirectional RNNs to construct a frequency domain model for predicting the power spectrum of buffeting response. The findings revealed that the trained neural network model exhibited higher prediction accuracy and reduced computational time compared to traditional buffeting analysis methods. Furthermore, Wang *et al.* (2020) leveraged random forest,

gaussian process regression, and LSTM models to forecast pulsating wind speed. The results demonstrated the model's high accuracy and generalization capability. These studies collectively highlight the ability of LSTM to perform rapid and precise prediction and modeling of unsteady flows.

The Reynolds number (Re) is a dimensionless parameter that quantifies the ratio of the inertial force to the viscous force within a fluid. Its impact on the aerodynamic characteristics of structures is intricate, with the predominance of either the viscous force or the inertial force depending on the magnitude of the Reynolds number. As the Reynolds number varies, diverse flow patterns manifest when wind interacts with structures (Niemann and Hölscher 1990, Li *et al.* 2014, Laima *et al.* 2018). Many studies have investigated Reynolds number effects. Thom *et al.* (1933) used arithmetic to determine the resistance and pressure distribution of flow around circular cylinder at $Re=20$. Achenbach *et al.* (1968) measured the drag coefficient and pressure distribution of flow around circular cylinder in the range of 6×10^4 to 5×10^6 in a wind tunnel. Jiang *et al.* (2021) employed the large eddy simulation method to examine the aerodynamic characteristics of flow around circular cylinder in Reynolds number range of 400 to 3900.

Wind typically hits a structure at a specific angle that can be decomposed into angles of attack and angles of direction. The thickness of the shear layer that separates after fluid flows around a structural section varies with different angles of attack. The aerodynamic characteristics of the structure are highly sensitive to angles of attack (Kozmar *et al.* 2012). Gao *et al.* (2017) conducted a wind tunnel test to measure the wind pressure around slotted and unslotted circular cylinders at angles of attack ranging from 0° to 90° . They also utilized particle image velocimetry technology to measure the wake characteristics at different angles of attack. The results provided a reference for flow control around slotted cylinders. Zhang *et al.* (2017) investigated vortex energy harvesting of square columns under different angles of attack. The study findings indicated that the angles of attack significantly influenced the vortex energy harvesting performance of a square column. Kim *et al.* (2014) analyzed the aerodynamic characteristics of the National Advisory Committee for Aeronautics (NACA) 0012 airfoil with the Reynolds number range of 2×10^4 to 5×10^4 and angles of attack range from -6° to 6° . The results further highlighted the sensitivity of structural aerodynamic characteristics to both angles of attack and Reynolds number.

Researchers have extensively investigated the aerodynamic characteristics of structures under various flow conditions. However, obtaining specific aerodynamic characteristics for particular structures in specific flow fields often necessitates time-consuming and costly wind tunnel tests or computational fluid dynamics simulations. Consequently, modeling the aerodynamic force coefficients of structures at different Reynolds numbers and angles of attack can enhance the efficiency of conventional methods and reduce calculation costs.

This study aims to develop a predictive model for structural unsteady aerodynamics using the LSTM network.

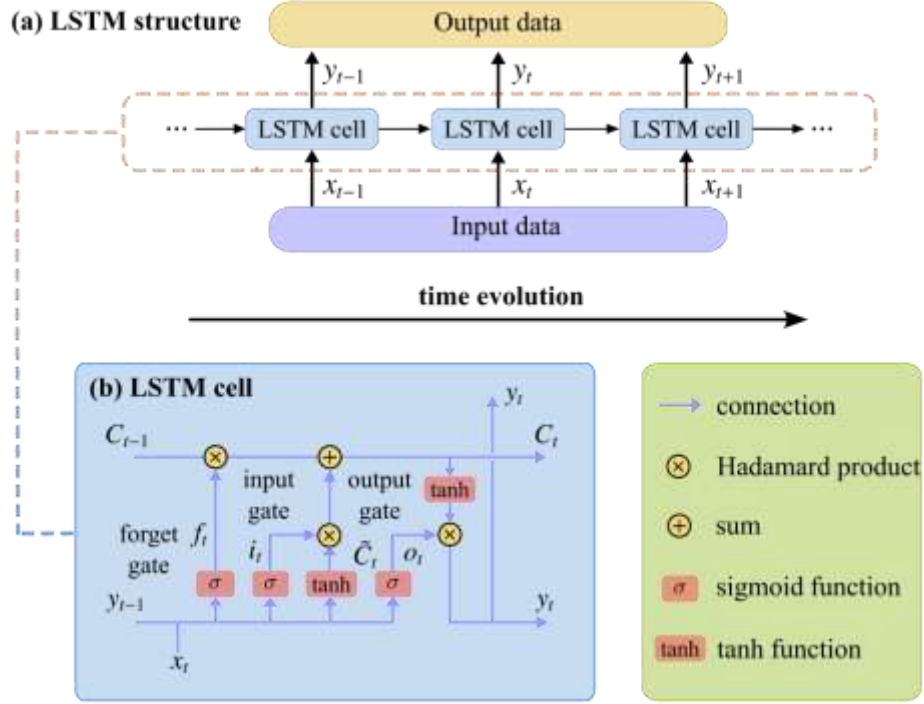


Fig. 1 (a) The structure of LSTM model (b) LSTM memory block in one cell

The LSTM model is trained and evaluated using numerical simulation results of structural unsteady aerodynamics under different Reynolds numbers and angles of attack. The accuracy and generalization performance of the model are validated through interpolation and extrapolation predictions of aerodynamic coefficients across various Reynolds numbers and angles of attack. Section 2 provides an overview of the LSTM model and the aerodynamic coefficient prediction methods employed in this paper. Section 3 presents the setup and results of the numerical simulation. Section 4 describes the dataset used for training and testing the LSTM model. The prediction results of the model are analyzed and discussed in Section 5. Finally, Section 6 provides a summary and concluding remarks.

2. Deep learning methods

2.1 Long short-term memory network model

Recurrent Neural Networks (RNNs) are capable of addressing time series prediction tasks that involve nonlinear input-output relationships by incorporating temporal dimensions into the network architecture. Nevertheless, when dealing with long input data sequences, RNNs encounter challenges in parameter updating through backpropagation, notably gradient vanishing and exploding. To mitigate these issues and enhance stability, Long Short-Term Memory (LSTM) networks are employed for predicting the nonlinear unsteady aerodynamics of structures in this study.

The LSTM model has three components: input, LSTM

cell and output, as shown in Fig. 1(a). The LSTM cell is a memory block that has three gates: input gate, forget gate and output gate. Each gate is regulated by an activation function, so that the gate value ranges from 0 to 1. The internal structure of the LSTM cell is shown in Fig. 1(b). The specific calculation formulas are as follows:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (1)$$

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (2)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (3)$$

$$C_t = f_t \otimes C_{t-1} + i_t \otimes \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (4)$$

$$h_t = o_t \otimes \tanh(C_t) \quad (5)$$

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad (6)$$

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \quad (7)$$

Where i_t , f_t and o_t represent input gate, forget gate and output gate, respectively. C_t represents the long-term memory. h_t represents the short-term memory, which belongs to long-term memory. W_i , W_f , W_o , and W_c are the weight matrixes of LSTM cell, respectively. b_i , b_f , b_o and b_c are the bias matrixes of LSTM cell, respectively. σ and $\tanh$ are active functions, which are used to achieve nonlinear transformation of input data. $\otimes$ indicates the

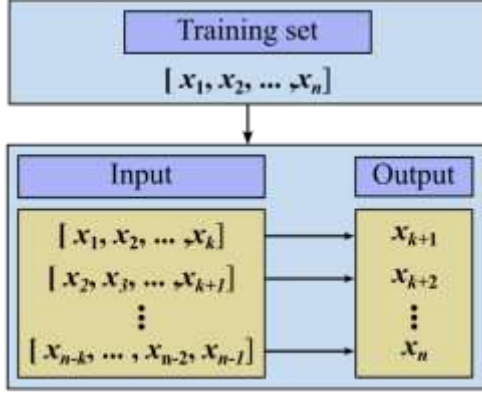


Fig. 2 Determination for the input and output variables

Hadamard product.

2.2 Methods for predicting aerodynamic characteristics

The training set of lift coefficients and drag coefficients is constructed using 1 to m sample points. The testing set of aerodynamic lift coefficients and drag coefficients is constructed based on m to n sample points. The relationship between input and output is shown in Fig. 2. The specific calculation formulas for the training set and the testing set are as follows:

$$X_{train}^L = \begin{bmatrix} C_L^1 & C_L^2 & C_L^3 & \dots & C_L^k \\ C_L^2 & C_L^3 & C_L^4 & \dots & C_L^{k+1} \\ \dots & \dots & \dots & \dots & \dots \\ C_L^{m-k} & C_L^{m-k+1} & C_L^{m-k+2} & \dots & C_L^{m-1} \end{bmatrix}_{(m-k) \times k} \quad (8)$$

$$Y_{train}^L = [C_L^{k+1} \ C_L^{k+2} \ C_L^{k+3} \ \dots \ C_L^m]_{1 \times (m-k)}^T \quad (9)$$

$$X_{train}^D = \begin{bmatrix} C_D^1 & C_D^2 & C_D^3 & \dots & C_D^k \\ C_D^2 & C_D^3 & C_D^4 & \dots & C_D^{k+1} \\ \dots & \dots & \dots & \dots & \dots \\ C_D^{m-k} & C_D^{m-k+1} & C_D^{m-k+2} & \dots & C_D^{m-1} \end{bmatrix}_{(m-k) \times k} \quad (10)$$

$$Y_{train}^D = [C_D^{k+1} \ C_D^{k+2} \ C_D^{k+3} \ \dots \ C_D^m]_{1 \times (m-k)}^T \quad (11)$$

$$X_{test}^L = \begin{bmatrix} C_L^{m+1} & C_L^{m+2} & C_L^{m+3} & \dots & C_L^{m+k} \\ C_L^{m+2} & C_L^{m+3} & C_L^{m+4} & \dots & C_L^{m+k+1} \\ \dots & \dots & \dots & \dots & \dots \\ C_L^{n-k} & C_L^{n-k+1} & C_L^{n-k+2} & \dots & C_L^{n-1} \end{bmatrix}_{(n-m-k) \times k} \quad (12)$$

$$X_{test}^D = \begin{bmatrix} C_D^{m+1} & C_D^{m+2} & C_D^{m+3} & \dots & C_D^{m+k} \\ C_D^{m+2} & C_D^{m+3} & C_D^{m+4} & \dots & C_D^{m+k+1} \\ \dots & \dots & \dots & \dots & \dots \\ C_D^{n-k} & C_D^{n-k+1} & C_D^{n-k+2} & \dots & C_D^{n-1} \end{bmatrix}_{(n-m-k) \times k} \quad (13)$$

where C_L^k and C_D^k represent the k^{th} sample point of the aerodynamic lift coefficients and aerodynamic drag coefficients, respectively. X_{train}^L and Y_{train}^L indicate the input and output of the aerodynamic lift coefficients when the LSTM model is trained, respectively. X_{train}^D and Y_{train}^D denote the input and output of the aerodynamic drag coefficients when the LSTM model is trained, respectively.

Table 1 Hyper-parameters of the LSTM network

Methods	Value
1 st LSTM layer cells	80
2 nd LSTM layer cells	120
LSTM input time-steps	10
Optimizer	Adam
Learning rate	0.001
Batch size	60
epochs	50

X_{test}^L and X_{test}^D express the testing set of lift coefficients and drag coefficients, respectively.

2.3 Training process of LSTM model

The framework of LSTM model is divided into five parts: numerical simulation, data selection, input data, model training and output data, as is shown in Fig. 3. Firstly, the numerical simulation is performed under different Reynolds number and angles of attack conditions. Secondly, the aerodynamic coefficients of 1 to n sample points is selected to form the dataset. Thirdly, the input dataset is constructed. Then, the structure of LSTM network and parameters are defined and the LSTM model is trained until the error requirements are met. When the model training loss is reduced to a certain value and there is no significant change in the loss as the number of epochs increases, the model training is stopped. Finally, the expected aerodynamic coefficients under different Reynolds numbers and angles of attack are output.

In order to ensure the accuracy and robustness of LSTM network prediction, the network structure shown in Fig. 4 was designed. The dropout layer is used to avoid overfitting. The dense layer guarantees that the output data is in the desired shape. The learning rate of the LSTM model is 0.001. After testing, the training loss of the model has satisfied the requirement when the number of training epochs is 50. The hyper-parameters of LSTM network are shown in Table 1.

2.4 Evaluation methods

Neural networks extensively rely on data for training purposes. However, this data cannot be utilized to evaluate the model's accuracy, similar to how students cannot use exercises they have repeatedly practiced as an exam paper since they already possess the answers. Therefore, in order to assess both the accuracy and generalization capability of the model, unseen data is introduced into the model for testing. The testing set is divided into two distinct components: interpolation prediction and extrapolation prediction. Interpolated predictions can be likened to exam questions based on the knowledge learned in class, while extrapolated predictions simulate exam questions that require applying knowledge beyond the classroom context to further evaluate proficiency.

The root mean square error (RMSE), mean absolute

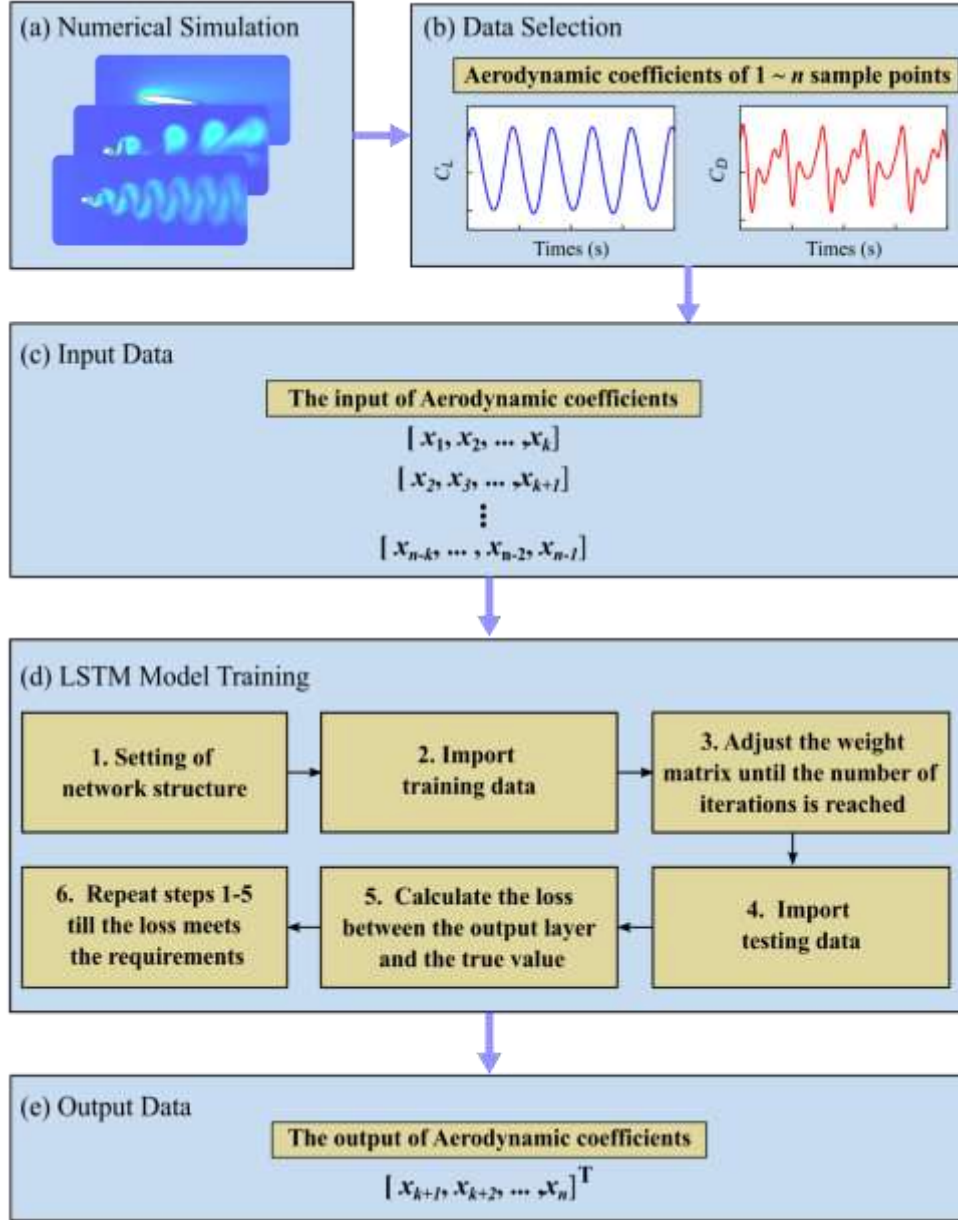


Fig. 3 The framework of the LSTM model. (a) Numerical simulation for flow field data. (b) Selection of aerodynamic coefficients. (c) The input dataset of LSTM model. (d) The training process of LSTM model. (e) output of aerodynamic coefficients

error (MAE), and mean relative error (MRE) are employed to quantitatively assess the accuracy of the model. The specific calculation formula are provided below:

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (y_i - f(x_i))^2} \quad (14)$$

$$MAE = \frac{1}{m} \sum_{i=1}^m |y_i - f(x_i)| \quad (15)$$

$$MRE = \frac{1}{m} \sum_{i=1}^m \frac{|y_i - f(x_i)|}{y_i} \quad (16)$$

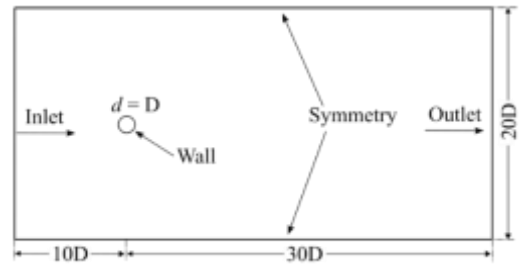


Fig. 5 Computational domain and boundary conditions of flow around a circular cylinder

where m , y_i and $f(x_i)$ represents the number of samples,

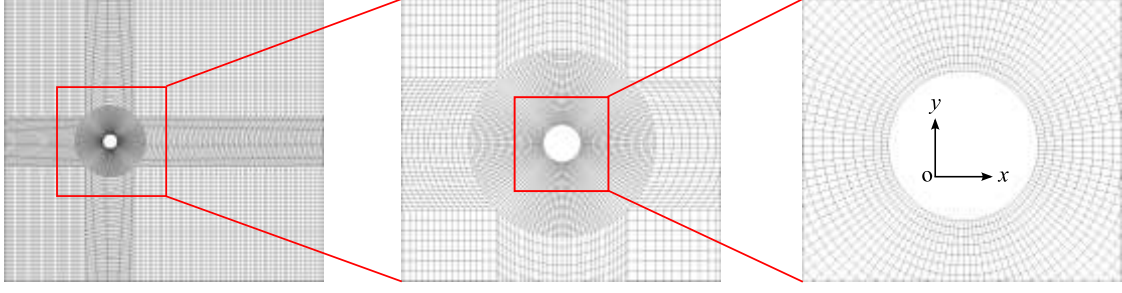


Fig. 6 Computational mesh for the OpenFOAM model of flow around a circular cylinder

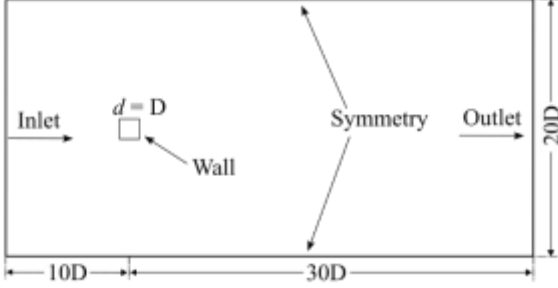


Fig. 7 Computational domain and boundary conditions of flow around a square cylinder

the real value and the predicted value, respectively.

3. Numerical simulation

3.1 Unsteady Reynolds Average Navier-Stokes method

The incompressible Navier-Stokes (N-S) equation is:

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (17)$$

$$\frac{\partial u_i}{\partial t} + \left(u_j \frac{\partial u_i}{\partial x_j} \right) = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_j^2} \quad (18)$$

where u_i is the i^{th} component of instantaneous velocity, p is instantaneous pressure, ρ is density, and ν is dynamic viscosity.

Time averaging of the N-S equation yields the unsteady Reynolds Average Navier-Stokes (URANS) equation:

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0 \quad (19)$$

$$\frac{\partial \bar{u}_i}{\partial t} + \left(u_j \frac{\partial \bar{u}_i}{\partial x_j} \right) = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\nu \frac{\partial^2 \bar{u}_i}{\partial x_j^2} - \overline{u_i u_j} \right) \quad (20)$$

From the formulas (18) and (20), it can be seen that compared with the original N-S equation, the URANS equation has an extra term which is named the Reynolds stress. A turbulence model needs to be required to enclose

the Reynolds stress term. The SST $k - \varepsilon$ turbulence model combines a free-flow $k - \varepsilon$ model with a near-wall $k - \varepsilon$ model. It is nearly a one-size-fits-all model in industrial applications. The SST $k - \varepsilon$ turbulence model is selected in this study due to its high accuracy and reliability.

3.2 Settings of different cases

The open-source software OpenFOAM is used to simulate three cases. To investigate the influence of Reynolds number on the LSTM model, URANS simulations are conducted for circular and square cylinders under varying Reynolds numbers. The LSTM model is utilized to predict the aerodynamic coefficients by employing interpolation and extrapolation techniques considering different Reynolds numbers. Furthermore, an additional investigation is carried out to explore the impact of angles of attack on the LSTM model. URANS simulation for the NACA0012 airfoil is performed with different angles of attack. Subsequently, the LSTM model is applied to predict the aerodynamic coefficients by considering the internal and external distribution in relation to the angles of attack.

3.2.1 Flow around a circular cylinder

The flow around a circular cylinder at Reynolds numbers of 100, 600 and 1000 are simulated, respectively. The calculation domain and boundary conditions are shown in Fig. 5. The diameter of the cylinder is D . The inlet of the flow field is defined as the velocity inlet and the flow velocity is 1m/s. The outlet of the flow field is defined as zero-pressure gradient outlet. The surface of the circular cylinder is a no-slip wall. The upper and lower boundaries are symmetrical. The mesh has about 20 thousand grids. The schematic of mesh is shown in Fig. 6.

3.2.3 NACA0012 airfoil

The structural geometry is changed from a blunt structure to a streamlined NACA0012 airfoil to further examine the generalization performance of the LSTM model. Based on the distribution of angles of attack, the NACA0012 airfoil with attack angles of 2° , 4° and 6° are simulated, respectively. The calculation domain and boundary conditions are shown in Fig. 9. The chord length of NACA0012 airfoil is 1m. The inlet of the flow field is defined as the velocity inlet and the flow velocity is 2 m/s. The dynamic viscosity coefficient is $1 \times 10^{-6} \text{ N}\cdot\text{s}/\text{m}^2$. The

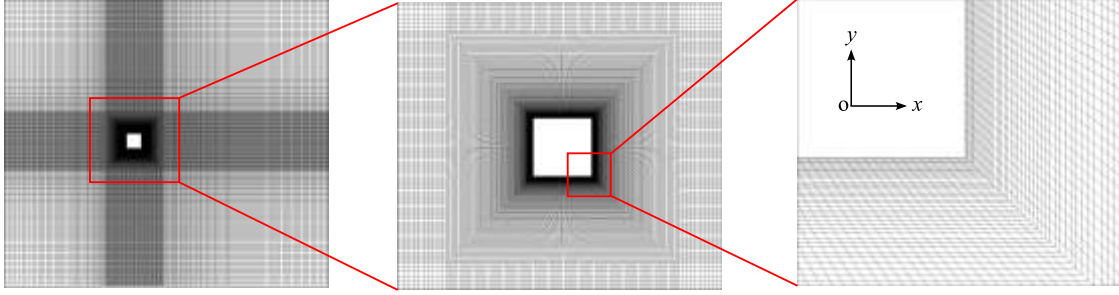


Fig. 8 Computational mesh for the OpenFOAM model of flow around a square cylinder

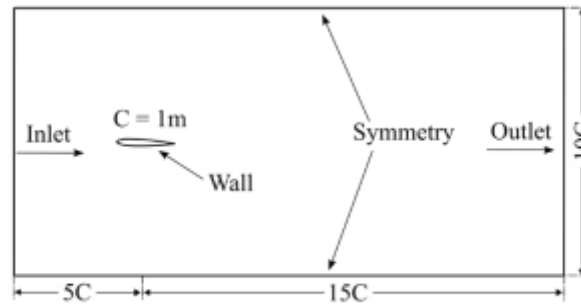


Fig. 9 Computational domain and boundary conditions of flow around NACA0012 airfoil

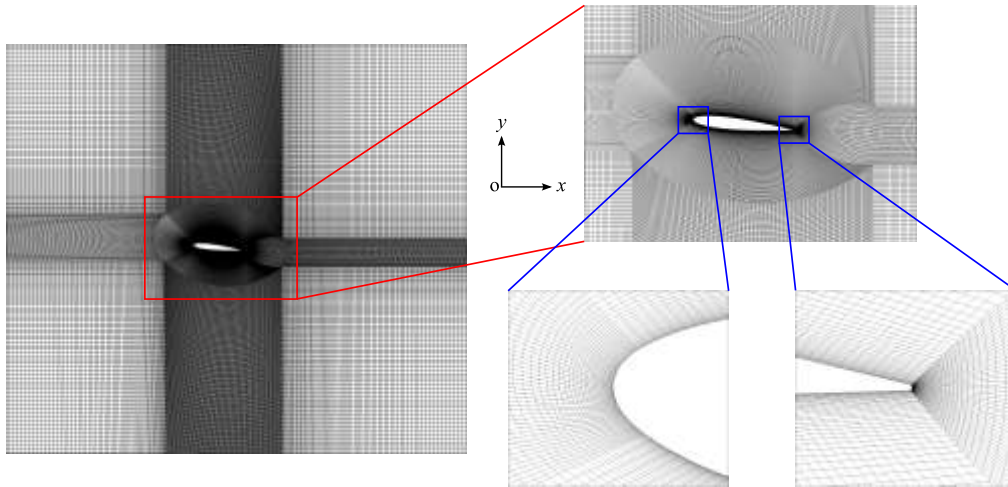


Fig. 10 Computational mesh for the OpenFOAM model of flow around NACA0012 airfoil

outlet is defined as zero-pressure gradient outlet. The airfoil surface is a no-slip wall. The upper and lower boundaries are symmetrical. The number of grids is about 120 thousand. The schematic of mesh is shown in Fig. 10.

3.3 Results and verification

3.3.1 Flow around a circular cylinder

The contour of the turbulent kinetic energy (k) of flow around a circular cylinder with different Reynolds numbers is shown in Fig. 11. The contour indicates the variation in vortex flow. The vortex structures around the circular cylinder reveal that the turbulence intensity is higher near

the cylinder. The wake vortex shedding position and the vortex structure vary significantly under different Reynolds numbers. Therefore, predicting and analyzing the aerodynamics of flow around a circular cylinder under different Reynolds numbers can help to understand the flow mechanism better and prevent wind-induced vibration damage of cylindrical structures.

3.3.2 Flow around a square cylinder

The contour of k of flow around a square cylinder with different Reynolds numbers is shown in Fig. 12. The contour of k can visually display the vortex changes when the wind passes the section of the square cylinder. The

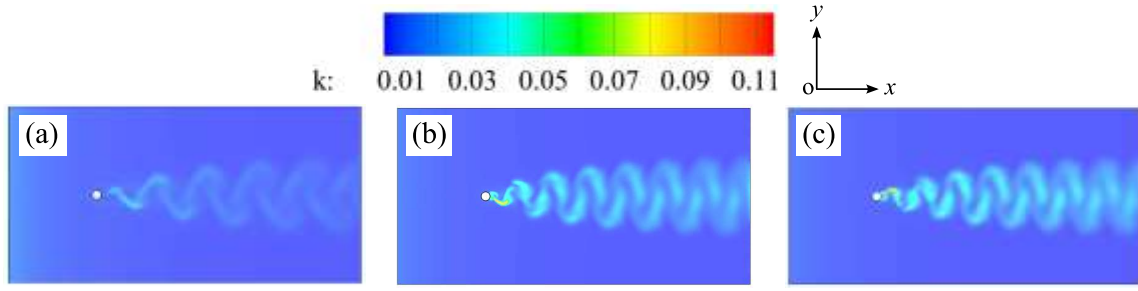


Fig. 11 The contour of the turbulent kinetic energy k of flow around a circular cylinder. (a) $Re=100$ (b) $Re=600$ (c) $Re=1000$

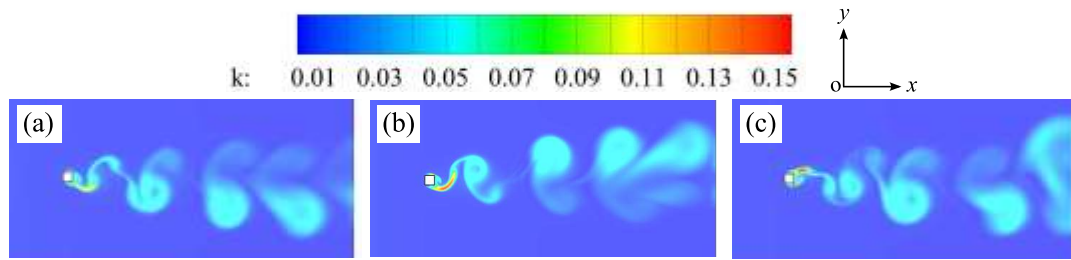


Fig. 12 The contour of the turbulent kinetic energy k of flow around a square cylinder. (a) $Re=3000$ (b) $Re=5000$ (c) $Re=7000$

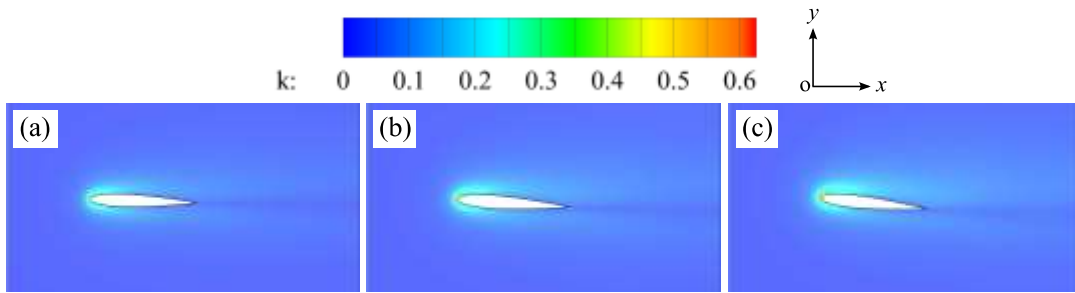


Fig. 13 The contour of the turbulent kinetic energy k of flow around NACA0012 airfoil. (a) the angle of attack is 2° (b) the angle of attack is 4° (c) the angle of attack is 6°

vortex structures around the square cylinder indicate higher turbulence near the square cylinder section. The wake vortex structure varies significantly under different Reynolds numbers. By comparing it with the vortex structure of flow around a circular cylinder, it can be seen that the vortex of wind around the structure changes considerably with the structural geometry and Reynolds number.

3.3.3 NACA0012 airfoil

The contour of k of NACA0012 airfoil at different angles of attack is shown in Fig. 13. The contour of k can display the vortex changes as the wind passes the airfoil. The vortex around the airfoil is not obvious due to the large Reynolds number, but the vortex structure around the airfoil indicates higher turbulence intensity at the front end of the airfoil. The k of the airfoil varies at different angles of attack.

Through a visual comparison of Fig. 11 and Fig. 13, it is evident that the turbulence around the structures is more intense. The vortex patterns generated by the wind passing

through the structures exhibit significant variations with changes in Reynolds number and angles of attack. Hence, the prediction of unsteady aerodynamic forces for structures subjected to varying Reynolds numbers and wind attack angles is crucial for gaining insights into the underlying flow mechanisms and ensuring the overall stability of the structures.

3.3.4 Verification of Aerodynamic coefficients

In order to ensure the reliability of the numerical simulation method employed in this study, the obtained results for the aerodynamic drag coefficients are compared and validated. The verification outcomes are presented in Table 2. The drag coefficients for flow around a circular cylinder at $Re = 100$ are compared with the findings of Stringer *et al.* (2014), revealing a relative error of 1.5%. For flow around a square cylinder at $Re = 5000$, a comparison with the study conducted by Norberg *et al.* (1993) yields a relative error of 3.8%. Furthermore, when examining flow around the NACA0012 airfoil at an angle of attack of 2° , the relative error is found to be 1.0% in comparison to the

Table 2 Verification of aerodynamic coefficients

Geometry	Type	Data source	CD	relative error
circular cylinder	Re=100	This paper	1.382	1.5%
		Stringer <i>et al.</i> (2014)	1.403	
square cylinder	Re=5000	This paper	2.130	3.8%
		Norberg <i>et al.</i> (1993)	2.210	
airfoil	$\alpha = 4^\circ$	This paper	0.020	1.0%
		Shi <i>et al.</i> (2013)	0.022	

Table 3 Data set of flow around a circular cylinder

Predicted category	Training set	Testing set
interpolation	Re=100, 400s-450s	Re=600, 460s-500s
	Re=1000, 400s-450s	
extrapolation	Re=100, 400s-450s	Re=1000, 460s-500s
	Re=600, 400s-450s	

Table 4 Data set of flow around a square cylinder

Predicted category	Training set	Testing set
interpolation	Re=3000, 400s-450s	Re=5000, 460s-500s
	Re=7000, 400s-450s	
extrapolation	Re=3000, 400s-450s	Re=7000, 460s-500s
	Re=5000, 400s-450s	

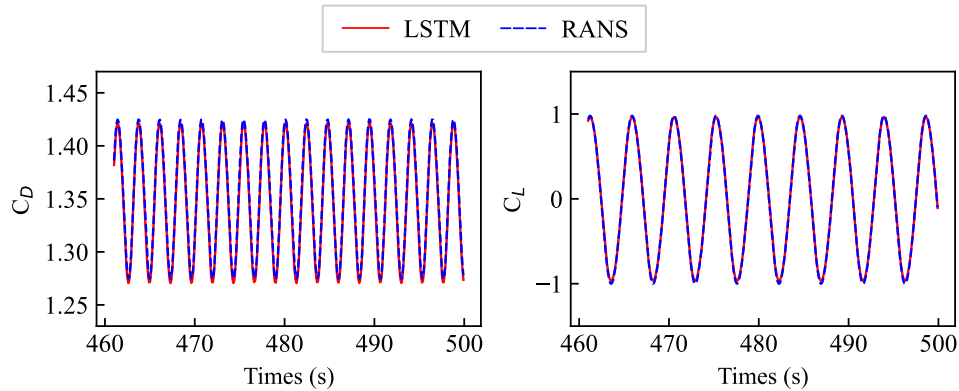


Fig. 14 The results of interpolation prediction of flow around a circular cylinder when the Reynolds number is 600

research performed by Shi *et al.* (2013). Notably, all the relative errors for the drag coefficients fall within 4%, thus confirming the accuracy of the numerical simulation method employed in this study (Tamura and Miyagi 1999, Yagmur *et al.* 2015).

4. Dataset using for LSTM model

4.1 Flow around a circular cylinder

In this study, the flow around a circular cylinder is utilized as a case to conduct interpolation and extrapolation predictions. The dataset pertaining to the flow around a

circular cylinder is presented in Table 3. The training set for the interpolation prediction comprises lift and drag coefficients obtained at Reynolds numbers of 100 and 1000. Conversely, the testing set comprises lift and drag coefficients acquired at a Reynolds number of 600. Regarding the extrapolation prediction, the training set encompasses lift and drag coefficients obtained at Reynolds numbers of 100 and 600, while the testing set encompasses lift and drag coefficients at a Reynolds number of 1000. The training set is formulated based on the URANS numerical simulation results spanning from 400s to 450s, encompassing a total of 980 sample points. On the other hand, the testing set is established using the URANS numerical simulation results from 460s to 500s, comprising

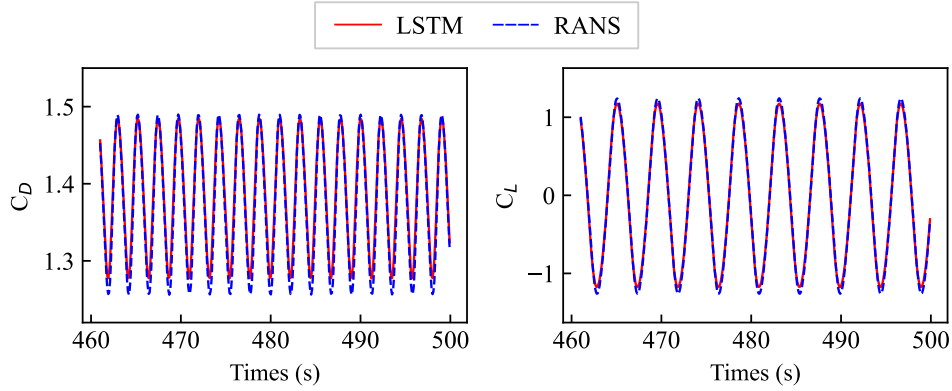


Fig. 15 The results of extrapolation prediction of flow around a circular cylinder when the Reynolds number is 1000

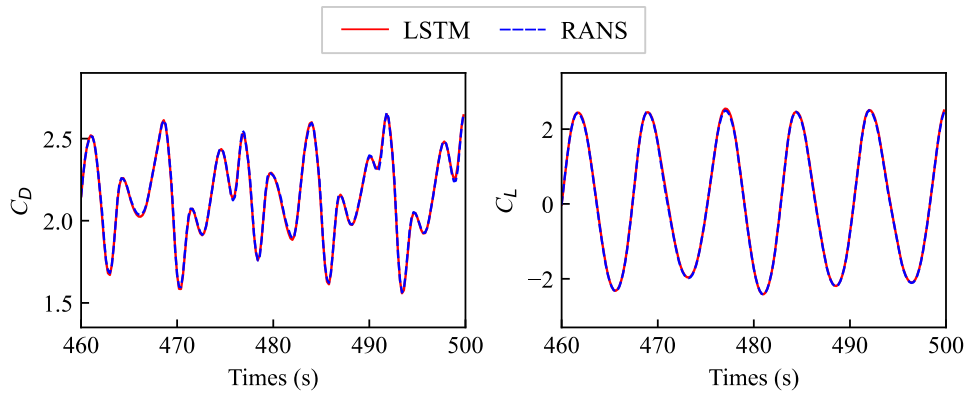


Fig. 16 The results of interpolation prediction of flow around a square cylinder when the Reynolds number is 5000

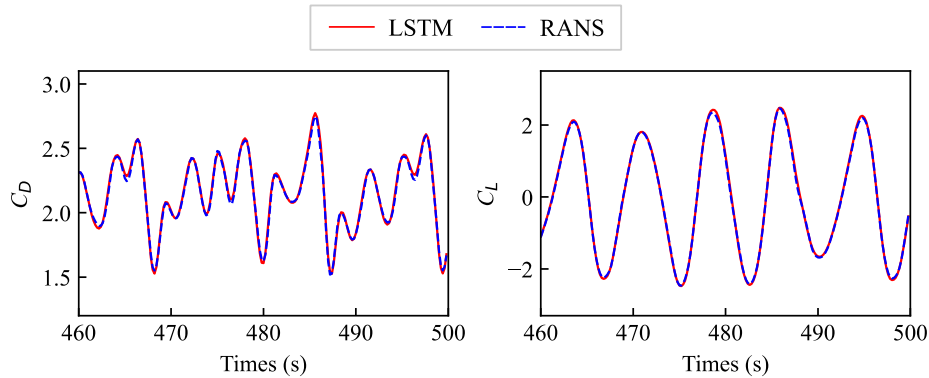


Fig. 17 The results of extrapolation prediction of flow around a square cylinder when the Reynolds number is 7000

a total of 390 sample points.

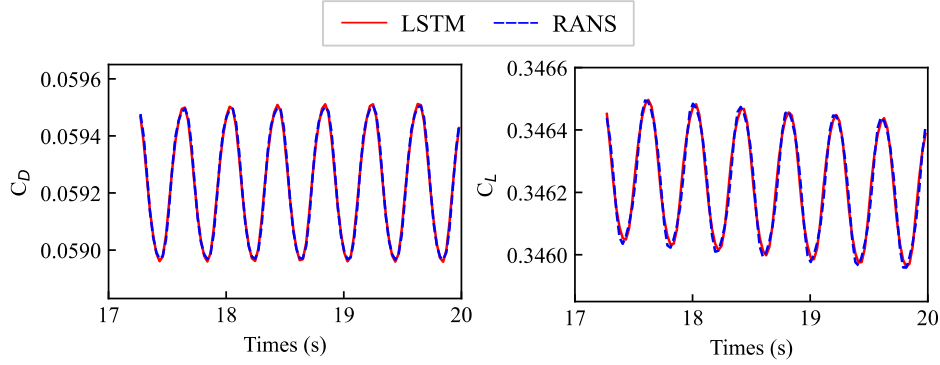
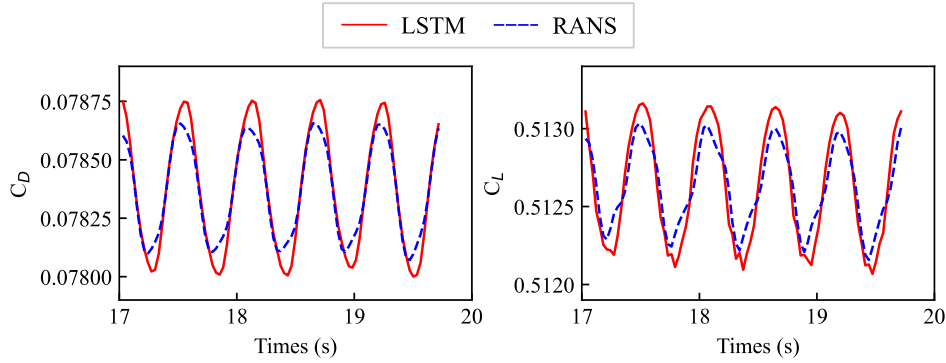
4.2 Flow around a square cylinder

The case of the flow around a square cylinder is performed interpolation and extrapolation predictions in this study. The dataset for interpolation and extrapolation of flow around a circular cylinder is shown in Table 4. The lift and drag coefficients at the Reynolds number of 3000 and 7000 are used for determining the training set of interpolation prediction, while the lift and drag coefficients

at the Reynolds number of 5000 are used for determining the testing set of interpolation prediction. The lift and drag coefficients at the Reynolds number of 3000 and 5000 are used for determining the training set of extrapolation prediction, while the lift and drag coefficients at the Reynolds number of 7000 are used for determining the testing set of extrapolation prediction. The results of URANS numerical simulation from 400s to 450s are employed for forming the training set, with a total of 480 sample points. The results of URANS numerical simulation from 460s to 500s are employed for forming the testing set,

Table 5 Data set of NACA0012 airfoil

Predicted category	Training set	Testing set
interpolation	$\alpha = 2^\circ$, 10s to 17s	$\alpha = 4^\circ$, 17s to 20s
	$\alpha = 6^\circ$, 10s to 17s	
extrapolation	$\alpha = 2^\circ$, 10s to 17s	$\alpha = 6^\circ$, 17s to 20s
	$\alpha = 4^\circ$, 10s to 17s	

Fig. 18 The results of interpolation prediction of NACA0012 airfoil when the angle of attack is 4° Fig. 19 The results of extrapolation prediction of NACA0012 airfoil when the angle of attack is 6°

with a total of 200 sample points.

4.3 NACA0012 airfoil

This study uses the case of NACA0012 airfoil to conduct interpolation and extrapolation predictions. The dataset of NACA0012 airfoil is shown in Table 5. The training set for interpolation prediction utilises the lift and drag coefficients at angles of attack (α) of 2° and 6° , while the testing set for interpolation utilises the lift and drag coefficients at angles of attack of 4° . The training set for extrapolation prediction utilises the lift and drag coefficients at angles of attack of 2° and 4° , while the testing set for extrapolation utilises the lift and drag coefficients at angles of attack of 6° . The results of URANS numerical simulation from 10s to 17s are chosen to obtain the training set, with a total of 450 sample points. The results of URANS numerical simulation from 17s to 20s are chosen to obtain the testing set, with a total of 100 sample points.

5. Results and discussions

5.1 Case of circular cylinder

Fig. 14 displays the outcomes of interpolation prediction for the aerodynamic lift coefficients (C_L) and the aerodynamic drag coefficients (C_D). Meanwhile, the results of extrapolation prediction are depicted in Fig. 15. It is noticeable that the prediction outcomes of the LSTM model exhibit excellent agreement with the trajectory of the URANS numerical simulation results over time. Additionally, there is a good match between the amplitude and period of the true and predicted values. Consequently, it can be concluded that the LSTM model can effectively predict the aerodynamic coefficients of the flow around a circular cylinder at varying Reynolds numbers.

5.2 Case of square cylinder

Fig. 16 shows the results of the interpolation prediction

Table 6 Prediction error of LSTM model

geometry	pattern	Re	angle	coefficient	RMSE	MAE	MRE
circular cylinder	interpolation	600	$\alpha = 0^\circ$	C_L	0.015	0.011	3.7%
				C_D	0.004	0.004	0.3%
	extrapolation	1000	$\alpha = 0^\circ$	C_L	0.038	0.032	4.0%
				C_D	0.014	0.011	0.8%
square cylinder	interpolation	5000	$\alpha = 0^\circ$	C_L	0.022	0.017	4.2%
				C_D	0.010	0.007	0.3%
	extrapolation	7000	$\alpha = 0^\circ$	C_L	0.042	0.032	5.9%
				C_D	0.019	0.015	0.7%
NACA0012	interpolation	2×10^6	$\alpha = 4^\circ$	C_L	$2.1e-5$	$1.8e-5$	0.01%
				C_D	$1.2e-5$	$1.0e-5$	0.02%
	extrapolation	2×10^6	$\alpha = 6^\circ$	C_L	$1.4e-4$	$1.2e-4$	0.02%
				C_D	$7.2e-5$	$5.4e-5$	0.07%

of flow around a square cylinder, while Fig. 17 shows the results of the extrapolation prediction. Compared with the case of circular cylinder, the case of square cylinder changes the structural geometry and increases the Reynolds number range. The results indicate that the trajectory, amplitude and period between the LSTM model results and the URANS simulation results are captured accurately. However, at the inflection points of the aerodynamic force change, the ability of the extrapolation prediction model is worse than the interpolation prediction model. Therefore, the LSTM model can also accurately predict the aerodynamic coefficients of square cylinder with different Reynolds number, but the prediction accuracy of the extrapolation model is lower than the interpolation model.

5.3 Case of NACA0012 airfoil

To evaluate the generalization performance of the LSTM model, predictions were conducted for the aerodynamic coefficients of the NACA0012 airfoil at various angles of attack. The interpolation prediction results are presented in Fig. 18, while the extrapolation prediction results are shown in Fig. 19. Examining the interpolation prediction, it can be observed that the LSTM model effectively captures the trajectory, amplitude, and period of the aerodynamic coefficients. Similarly, for the extrapolation prediction, the LSTM model can broadly capture the trajectory, amplitude, and period of the aerodynamic coefficients, though the error in the extrapolation prediction is slightly higher than the interpolation prediction at the inflection points of the aerodynamic coefficients. Overall, the LSTM model demonstrates accurate prediction capabilities for the aerodynamic coefficients of the NACA0012 airfoil at different angles of attack. However, it should be noted that the prediction accuracy of the extrapolation is lower compared to the interpolation.

5.4 Prediction error analysis

The prediction error of LSTM model is calculated from

Eqs. (14)-(16), as shown in Table 6. For the case of circular cylinder, the MRE of interpolation prediction of the lift coefficients and drag coefficients is 3.7% and 0.3%, respectively. The MRE of extrapolation prediction of the lift coefficients and drag coefficients is 4.0% and 0.8%. For the case of square cylinder, the MRE of interpolation prediction of the lift coefficients and drag coefficients is 4.2% and 0.3%, respectively. The MRE of extrapolation prediction of the lift coefficients and drag coefficients is 5.9% and 0.7%. For the case of NACA0012 airfoil, the MRE of interpolation prediction of the lift coefficients and drag coefficients is 0.01% and 0.02%, respectively. The MRE of extrapolation prediction of the lift coefficients and drag coefficients is 0.02% and 0.07%.

According to the report, the LSTM model exhibits the lowest prediction error in interpolation, while the error increases for extrapolation as the range of Reynolds numbers and angles of attack widens. The study demonstrates the successful reconstruction of unsteady aerodynamic forces for structures across various Reynolds numbers and angles of attack, thereby establishing the model's generalization capabilities.

6. Conclusions

In this paper, a LSTM neural network model is proposed for predicting unsteady aerodynamics in time-series. The model takes the aerodynamic coefficients of n sample points as input and predicts the aerodynamic coefficient of the $n+1$ sample point. To validate the model's performance, the URANS method is employed to calculate the lift and drag coefficients of three different geometries (i.e., circular cylinder, square cylinder, and NACA0012 airfoil) at various Reynolds numbers and angles of attack. The LSTM model is trained and tested on these aerodynamic coefficients data. The main conclusions from this study are as follows:

(1) The LSTM model exhibits high accuracy in predicting the trajectories, amplitudes, and periods of lift and drag coefficients over time. Comparisons with the URANS results demonstrate a strong agreement between

the prediction outcomes of the LSTM model and the actual aerodynamic coefficients. Additionally, the model demonstrates superior generalization capabilities in accurately predicting aerodynamic coefficients across various Reynolds numbers and angles of attack. Consequently, it can be relied upon to provide dependable predictions for the aerodynamic forces acting on engineering structures.

(2) The findings reveal that while the extrapolation errors are slightly higher than the interpolation errors, the prediction errors remain below 6% across all scenarios. Moreover, it is observed that the prediction errors escalate as the Reynolds number and angles of attack become more diverse, indicating a correlation between error and the widening generalization of these parameters.

(3) The training of the LSTM model on the generalization of Reynolds numbers is only performed in the subcritical partition in this paper. In the next study, the prediction of aerodynamic coefficients across Reynolds number partitions is considered to further improve the generalization of the LSTM model. The combination of LSTM and flow field reduced-order model is also considered to develop to capture the temporal evolution characteristics and predict future flow fields.

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